

Spooky Topics in Hydrology

EPS 522/ENVS 423L

November 6, 2025



WEBSTER WATERSHED RESILIENCE LAB

Join us!

We are hiring a **data manager** for a new research project: FOR-NM (Forest Research for New Mexico Water and Carbon Management).

- Data collection based in the Santa Fe Watershed and the Gila Wilderness, asking questions about forest management influences on water and carbon resources.
- The data manager will collate data across the project, including high-frequency sensor data, remote sensing products, and model outputs.
- Post-masters probably ideal
- Application open until filled



Housekeeping

- HW 7 due next Thursday
- No class next Tuesday
- Tech memo template posted to GitHub
- Presentation rubric posted to GitHub
- Tech memo rubric coming soon
- Lit. review feedback coming soon

Where did Earth's water come from?



Extrplanetary Sources?



Carbonaceous chondrite (up to 20% water)

Permafrost zombie virus



Dousing (water witching)

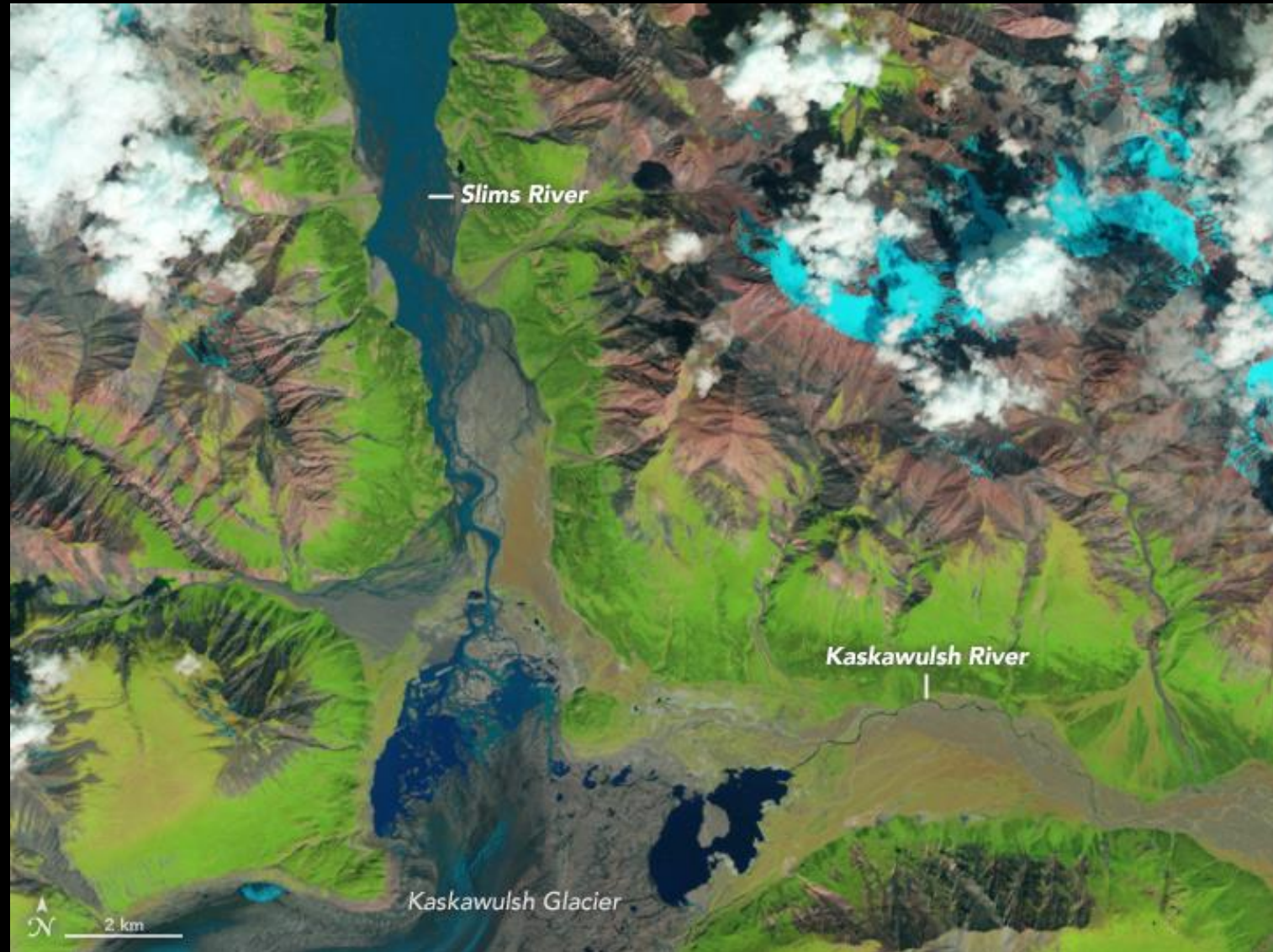


“Using a forked stick, rod, pendulum, or similar device to locate underground water, minerals, or other hidden or lost substances” (USGS)

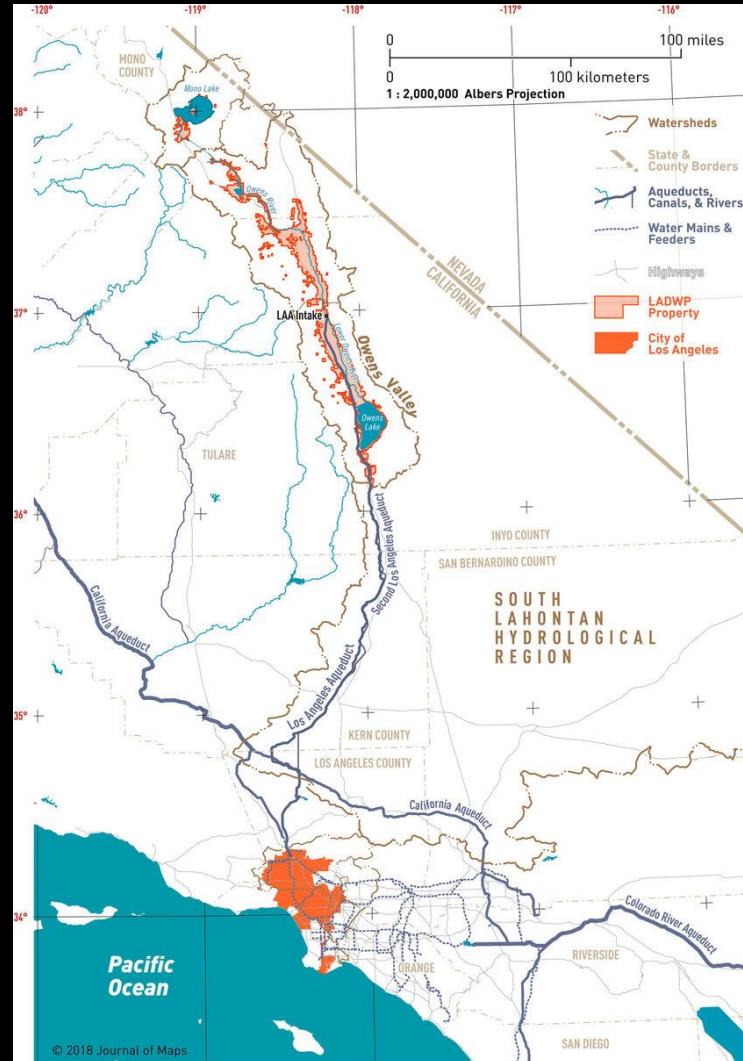
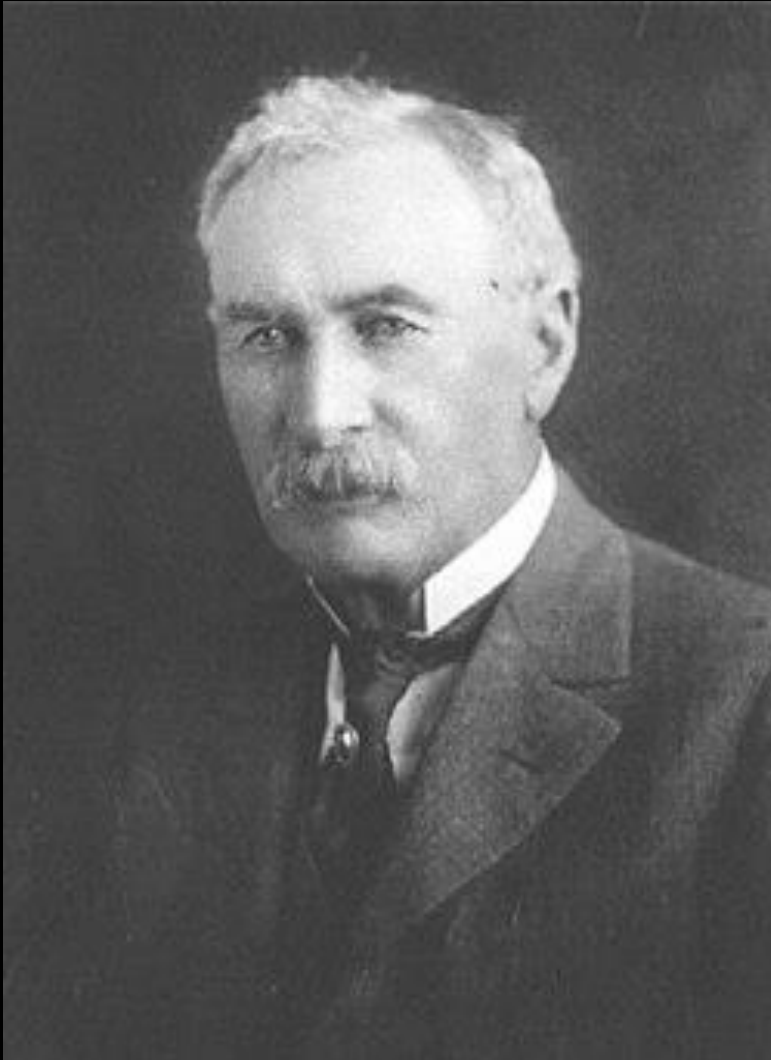
River Piracy

<https://earthobservatory.nasa.gov/images/90116/river-piracy-in-the-yukon>

<https://www.youtube.com/watch?v=4t1W0E1H8C0>



A different kind of river piracy...



In 1902, LA had a population of 200,000 and growing and was running out of water.

Mulholland began quietly buying up land with riparian water rights in the Owens Valley and constructed the 233-mile-long Los Angeles aqueduct in 1913.



By 1924, the agricultural economy in Owens Valley had been crippled, and the residents attempted to blow up the aqueduct near Lone Pine. This is referred to as the California Water Wars.

The Owens Lake was completely dry by 1926 and as LA kept growing, they began diverting water that fed into Mono Lake starting in 1941.



This threatened a delicate ecosystem in Mono Lake, as declining levels was impacting brine shrimp, whose presence attracts migratory birds.



The Mono Lake Committee sued LA Dept. of Water and Power, and the Supreme court ruled in their favor based on the Public Trust doctrine: “an affirmation of the duty of the state to protect the people’s common heritage of streams, lakes, marshlands and tidelands”

Nuclear Legacy



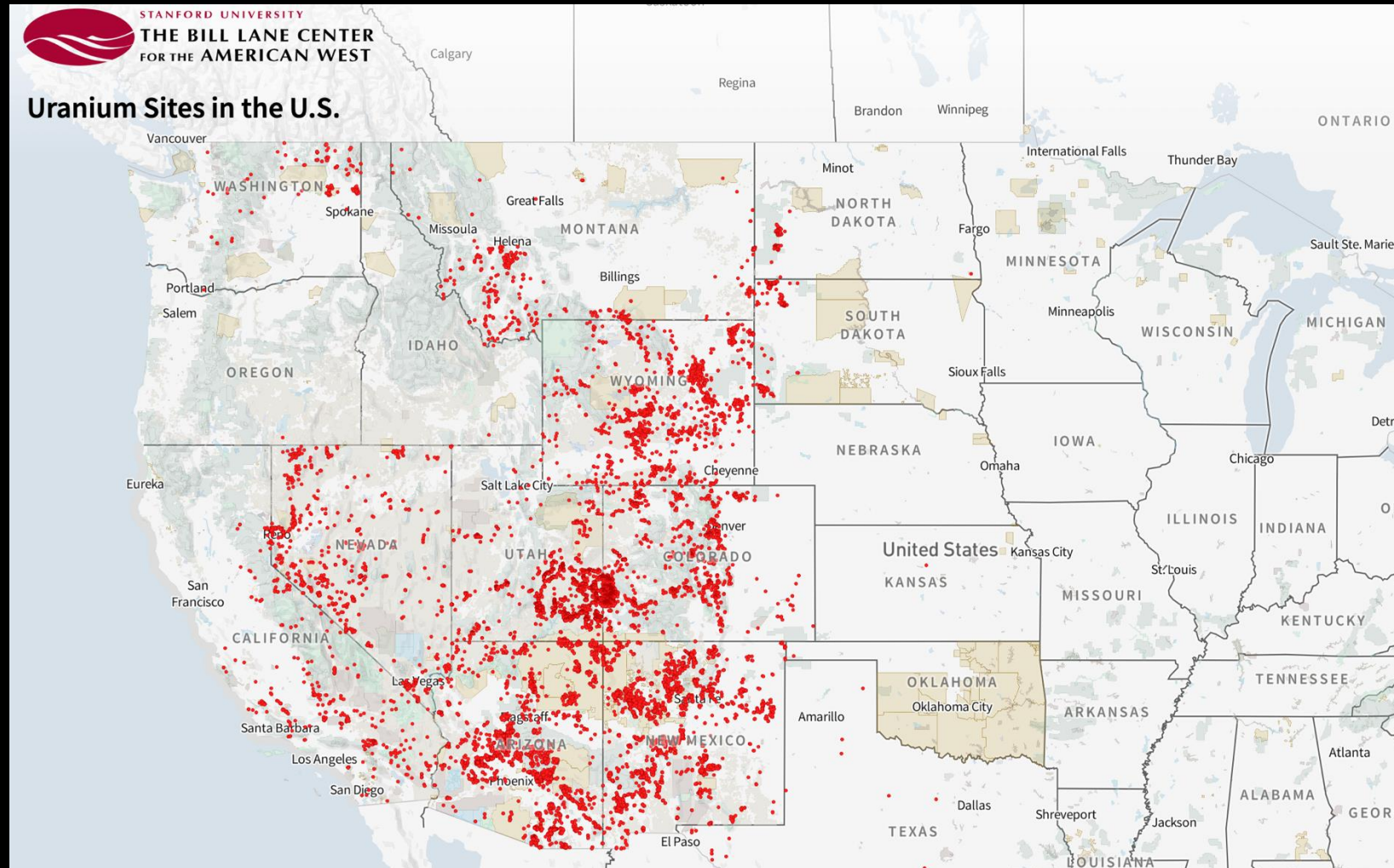
Manhattan Project

- R&D program during WWII to develop and produce the first nuclear weapons
- 1942-1946, \$24 billion in today's money, mostly spent on building and operating fissile material production plants
- Enriched uranium in Oak Ridge
- Enriched Plutonium at Hanford Site

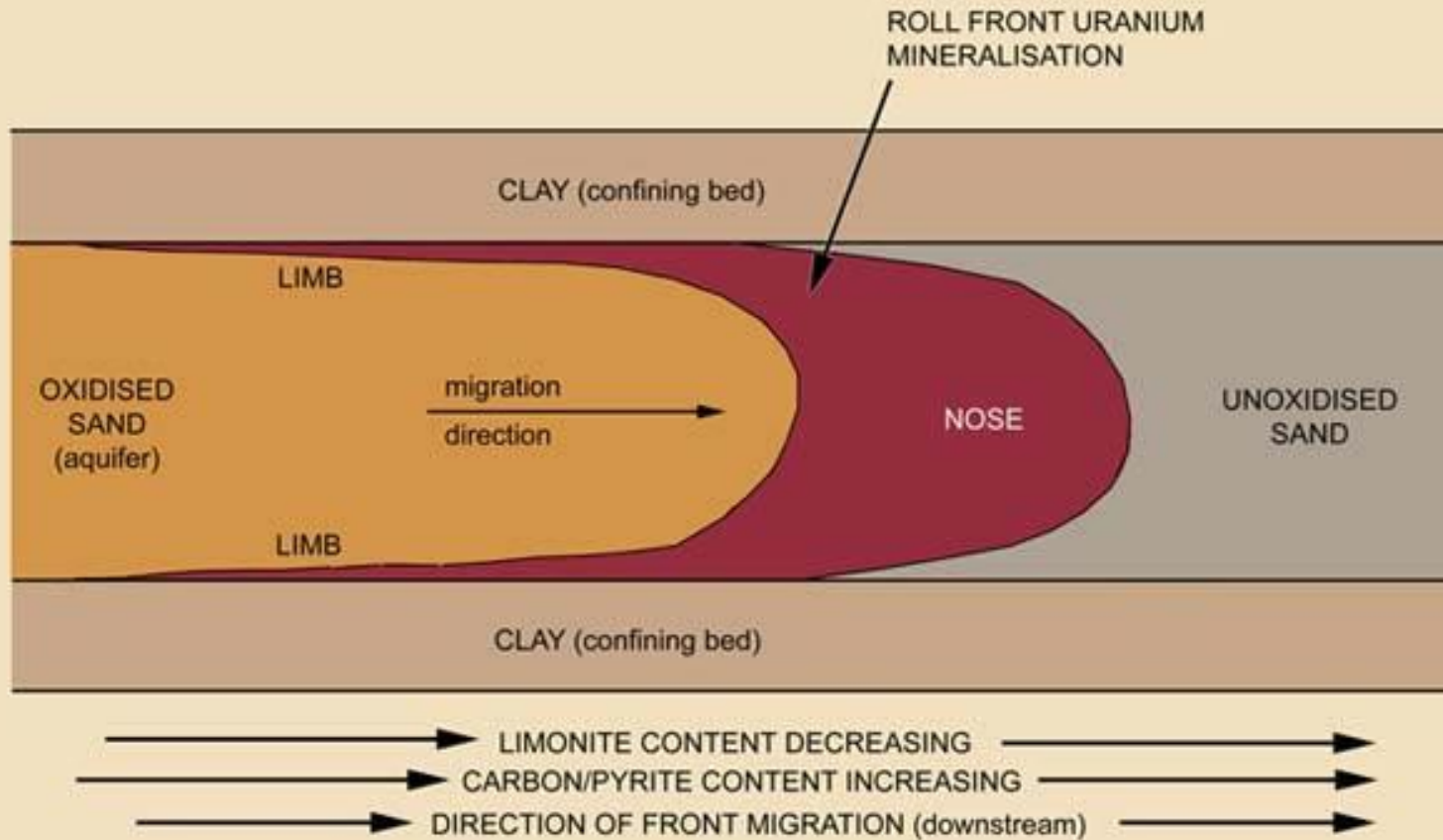


Uranium Mines

- Most US uranium mined for nuclear power
- Only 14% of ore used in Manhattan project came from Colorado Plateau (including New Mexico)



Roll-front Deposits



Uranium deposits formed as oxidized groundwater in sandstone aquifers (carrying the more soluble U[VI]) encounters reducing conditions (as a result of organic material in the host rock).

U(VI) reduces to the less soluble U(IV), which precipitates out of solution, forming a mineralized nose at the leading edge of flow.

ROLL FRONT URANIUM MODEL CROSS-SECTION

Adapted from published sources



Shinkolobwe Mine, DRC

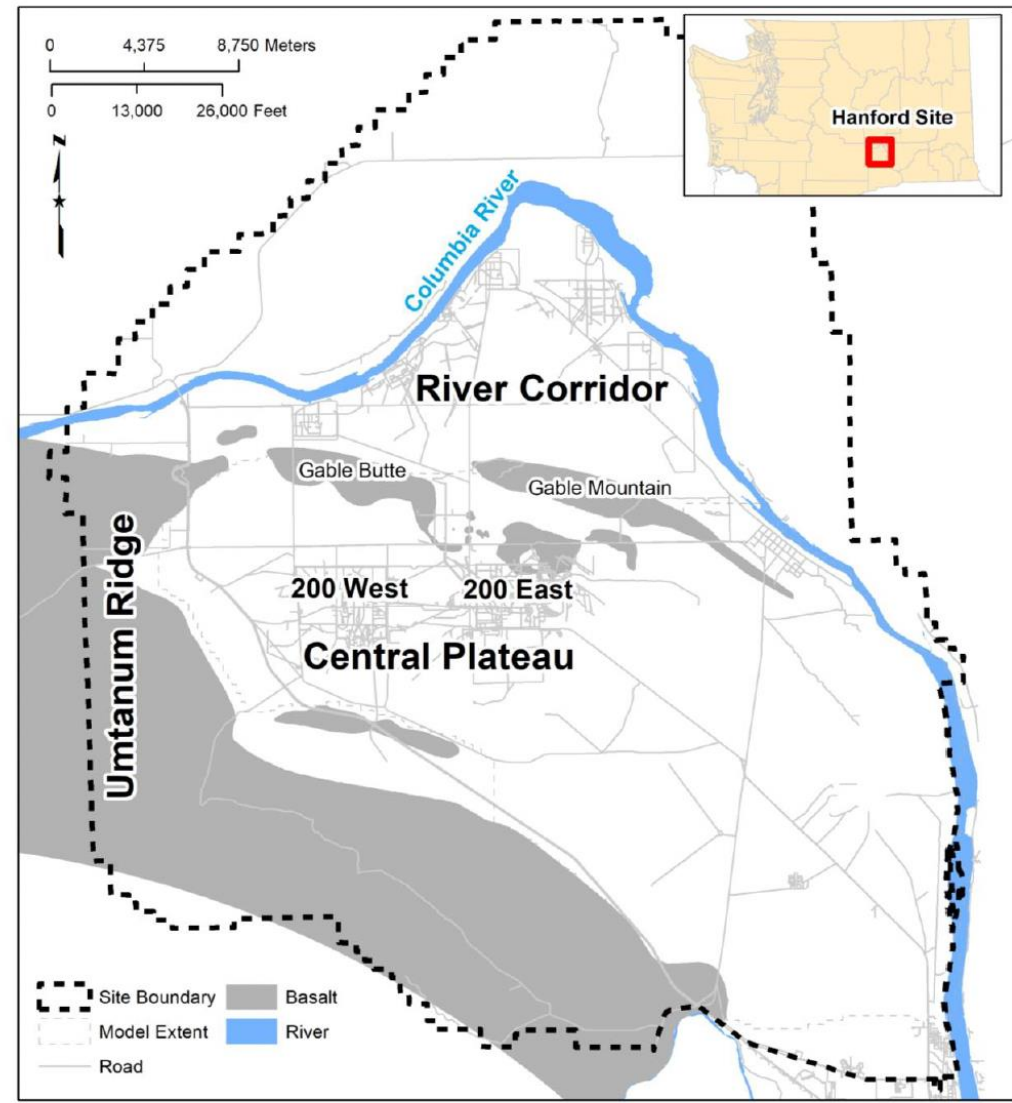


Shinkolobwe Mine



JFK and Mobutu Sese Seko

Plutonium Enrichment



Bomb testing



Trinity Test, July 16, 1945



Bikini Atoll testing, 1950s

Nevada Test Site

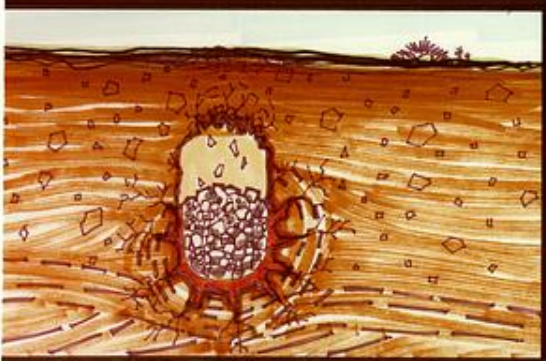
DETONATION SEQUENCE FOR AN UNDERGROUND NUCLEAR TEST



Underground, billions of atoms release energy through fusion or fission. Rock vaporizes, melts and fractures.



A spherical cavity forms. A layer of molten materials containing radioactive products forms, falls to the floor and solidifies.



As it cools, pressure inside the cavity decreases. Earth and loose rock fall into the cavity in a process called "chimneying."



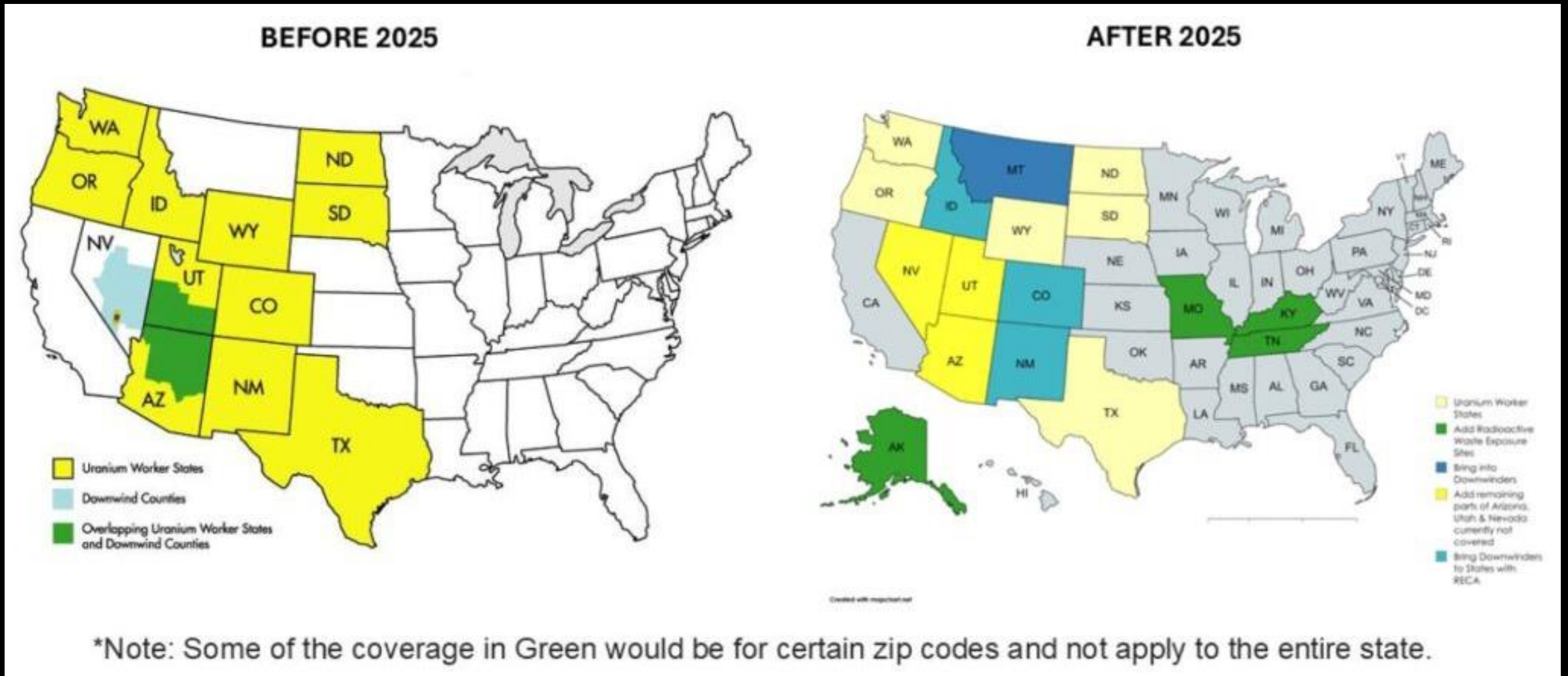
The earth usually sinks at the surface leaving a saucer-like subsidence.



The Aftermath

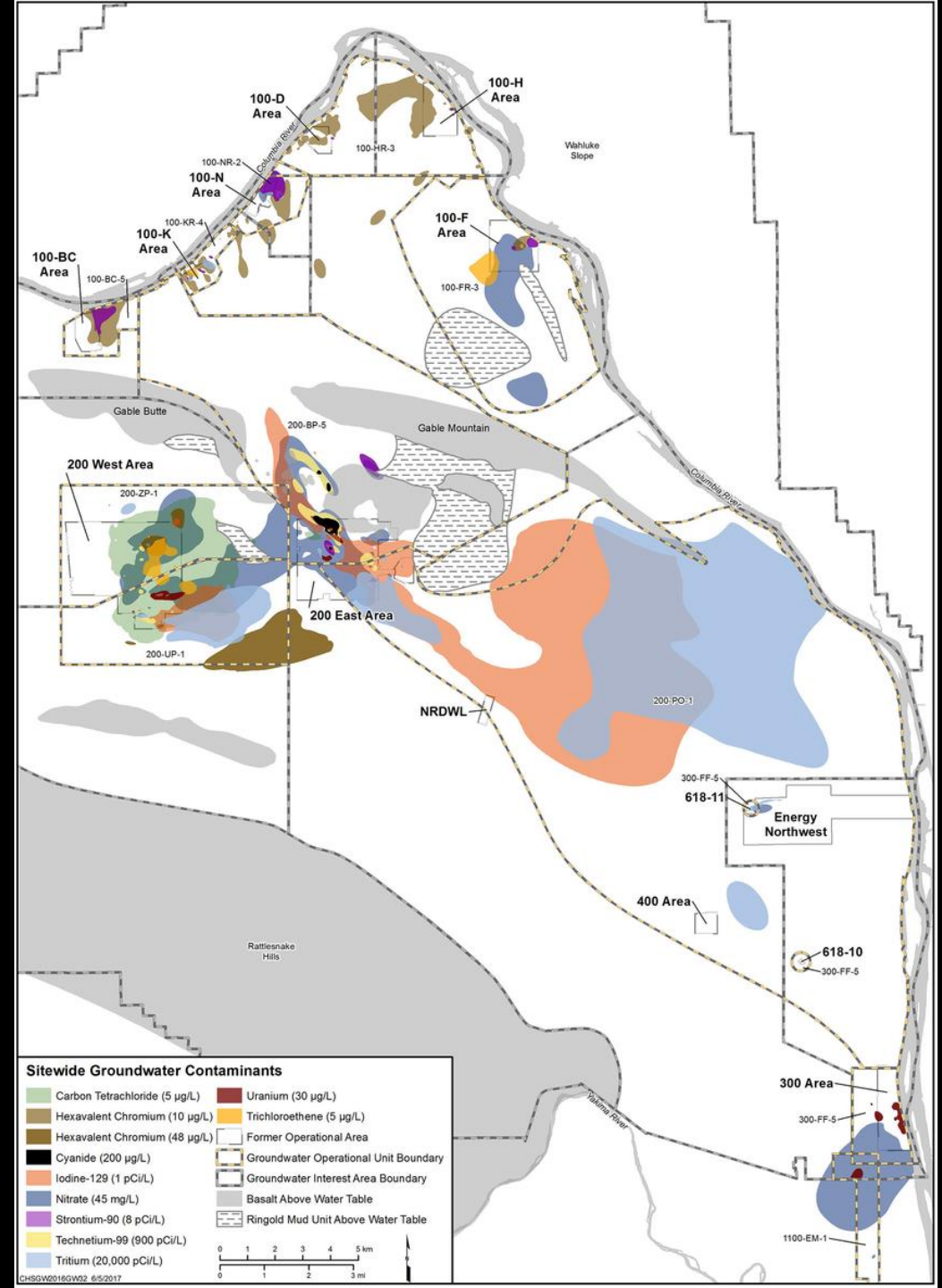
- 1989 formation of Dept. of Energy (DOE) Office of Environmental mgmt to cleanup the mess
 - Once a site is closed, responsibility is transferred to the Office of Legacy mgmt for long-term monitoring
 - Focus has shifted to cleanup (difficult to do given the extremely hazardous nature of the waste) and long-term disposal
- 1990 Radiation Exposure Compensation Act (RECA) passed to compensate “atomic veterans,” uranium workers, and downwinders.
- 1996 Treaty banning nuclear testing

RECA



Have to prove that you resided in the covered areas during specified durations of testing or live in zip codes affected by Manhattan project waste and contracted a “covered illness” (lots of types of cancers)

Hanford Site



These contaminants are long-lived...

Remediation Activities

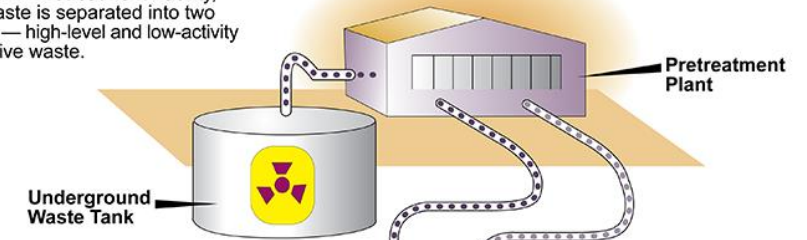


The Hanford Site has an annual operating cost of around \$5 billion per year, and is estimated to cost \$300-600 billion to complete cleanup

Vitrification Process

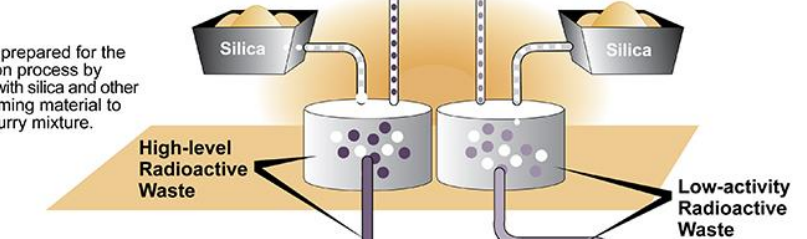
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At the WTP Pretreatment Facility, liquid waste is separated into two streams — high-level and low-activity radioactive waste.



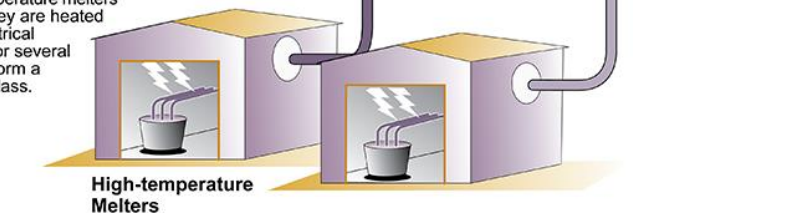
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Waste is prepared for the vitrification process by mixing it with silica and other glass-forming material to form a slurry mixture.



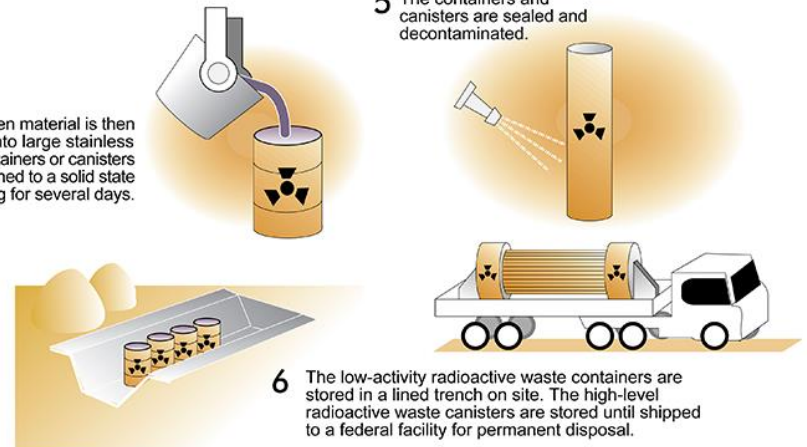
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The mixtures are fed into high-temperature melters where they are heated with electrical current for several days to form a molten glass.



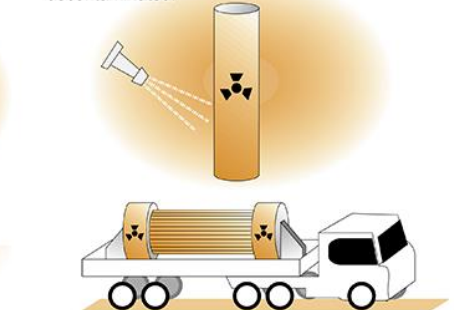
4

The molten material is then poured into large stainless steel containers or canisters and returned to a solid state by cooling for several days.



5

The containers and canisters are sealed and decontaminated.



6

The low-activity radioactive waste containers are stored in a lined trench on site. The high-level radioactive waste canisters are stored until shipped to a federal facility for permanent disposal.

Contaminant Treatment System

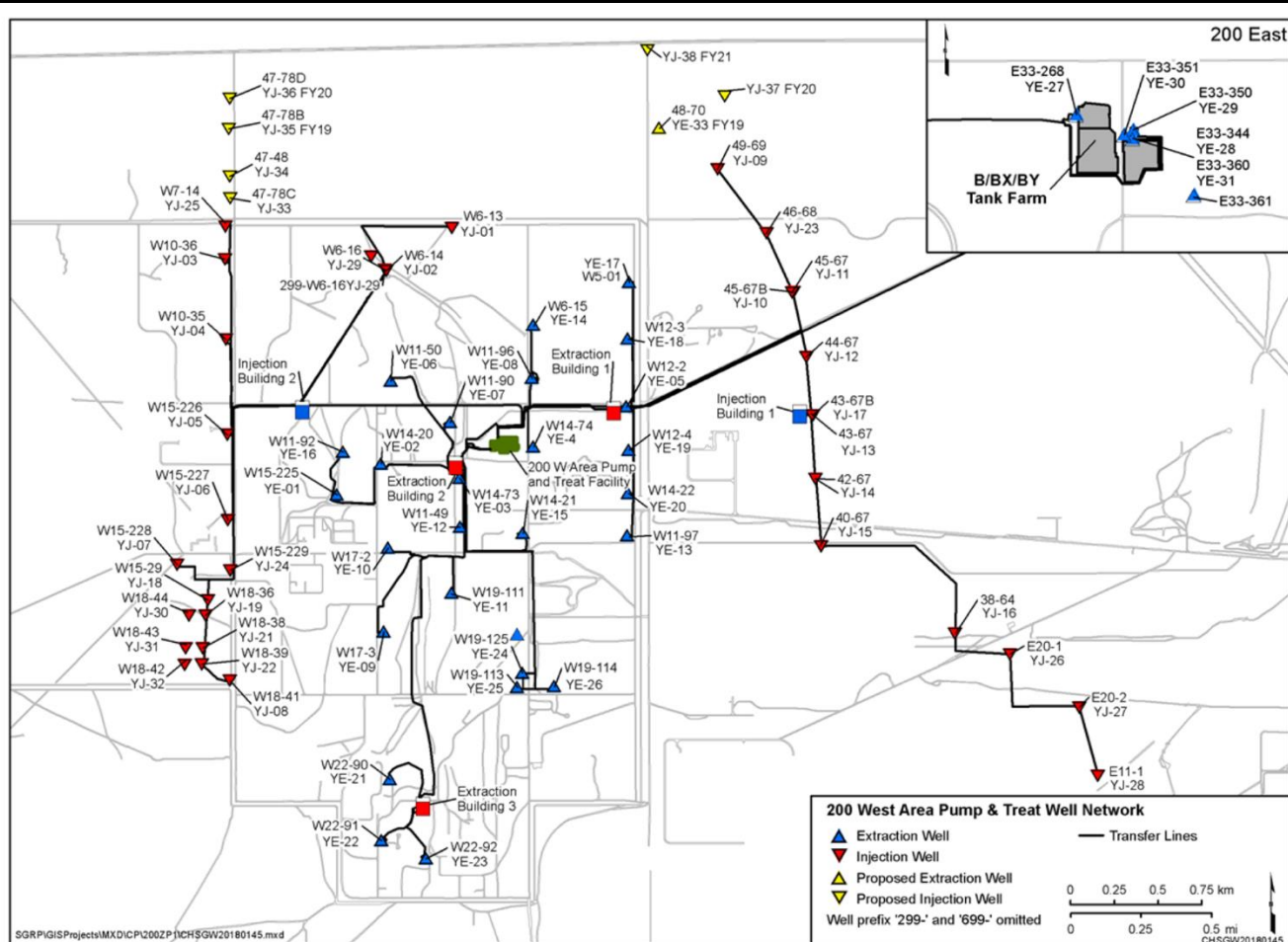


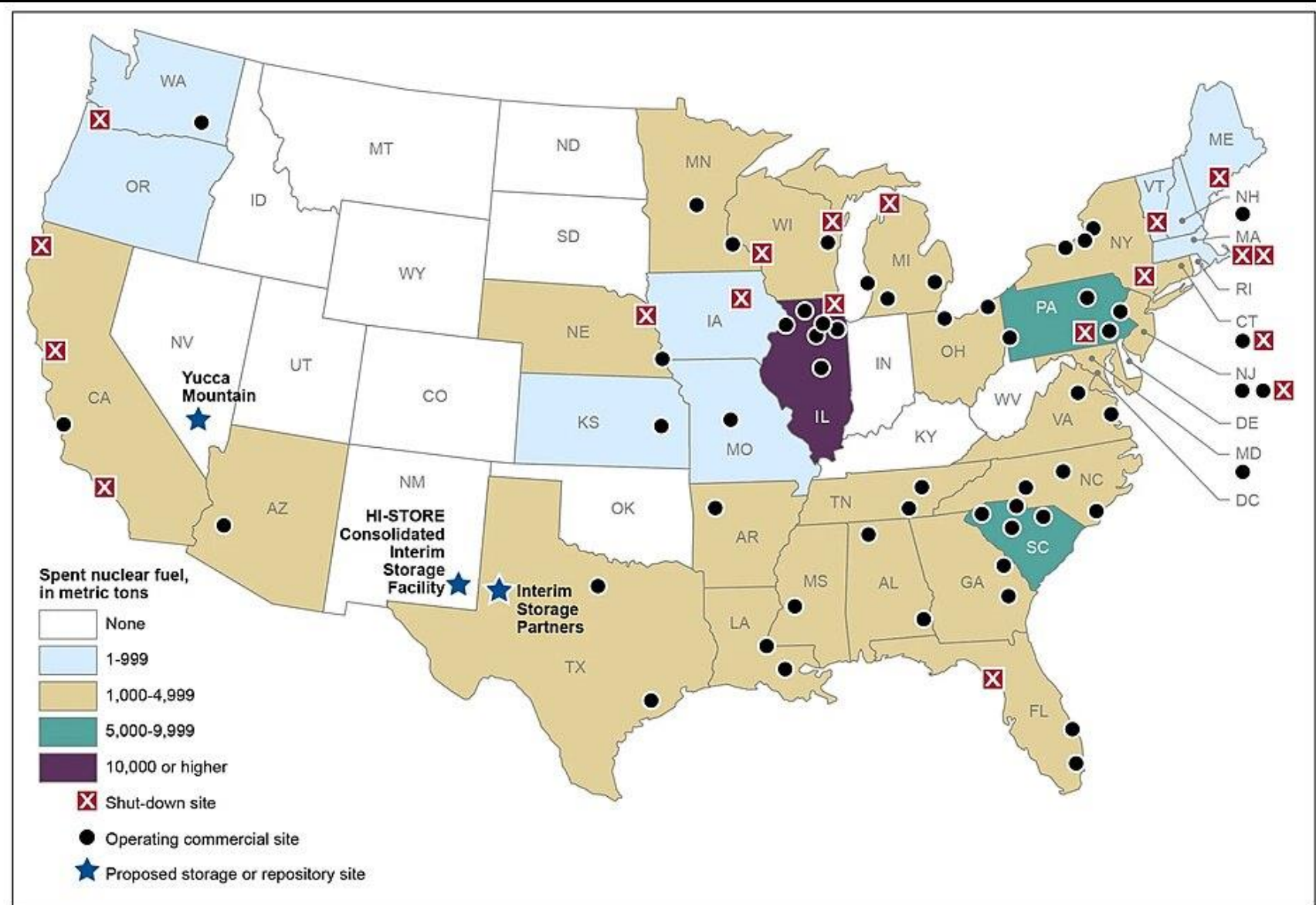
Figure 3. 200 West P&T Extraction and Injection Wells, September 2017



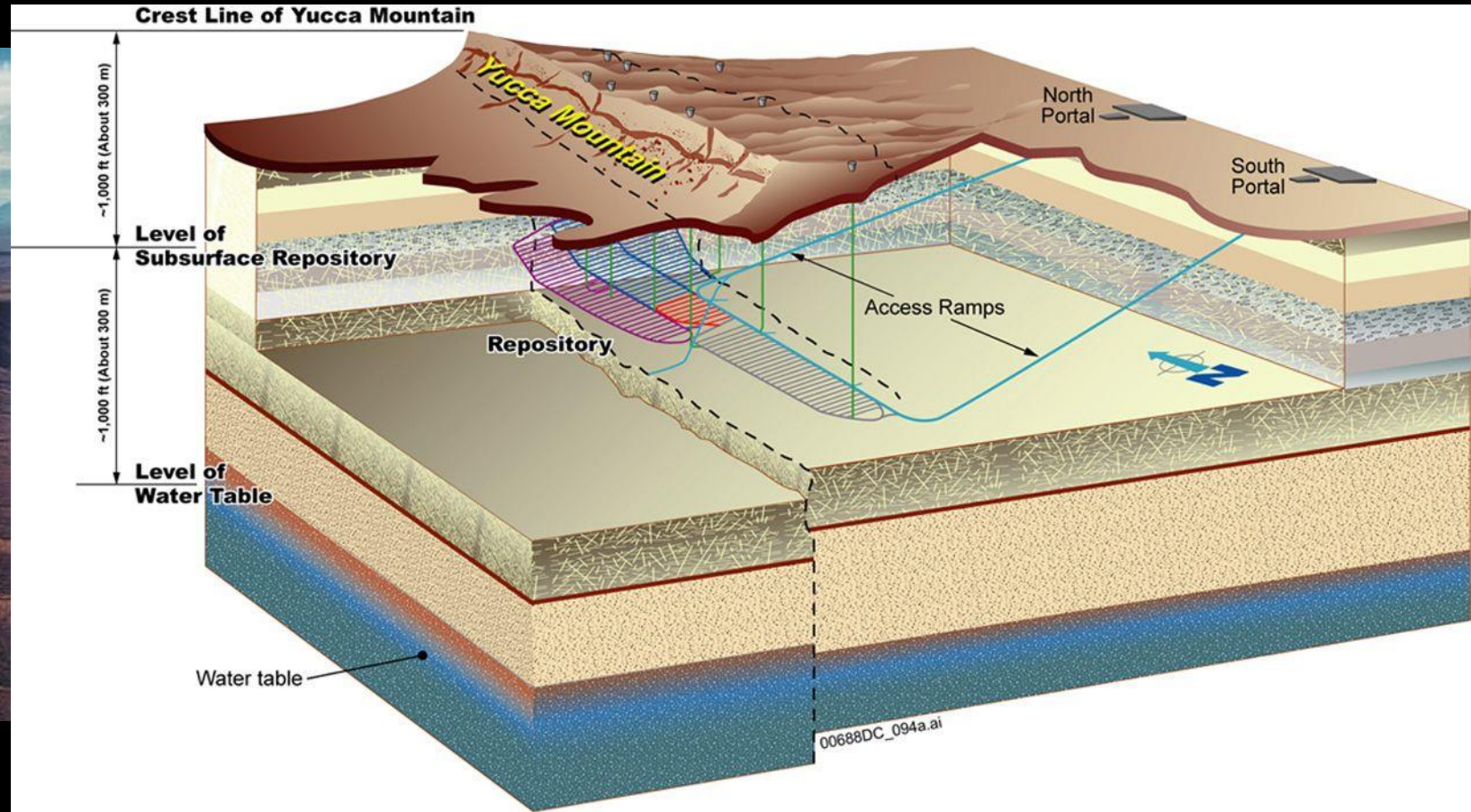
Federal contractors involved in Hanford cleanup



Long-term Radioactive Waste Disposal?

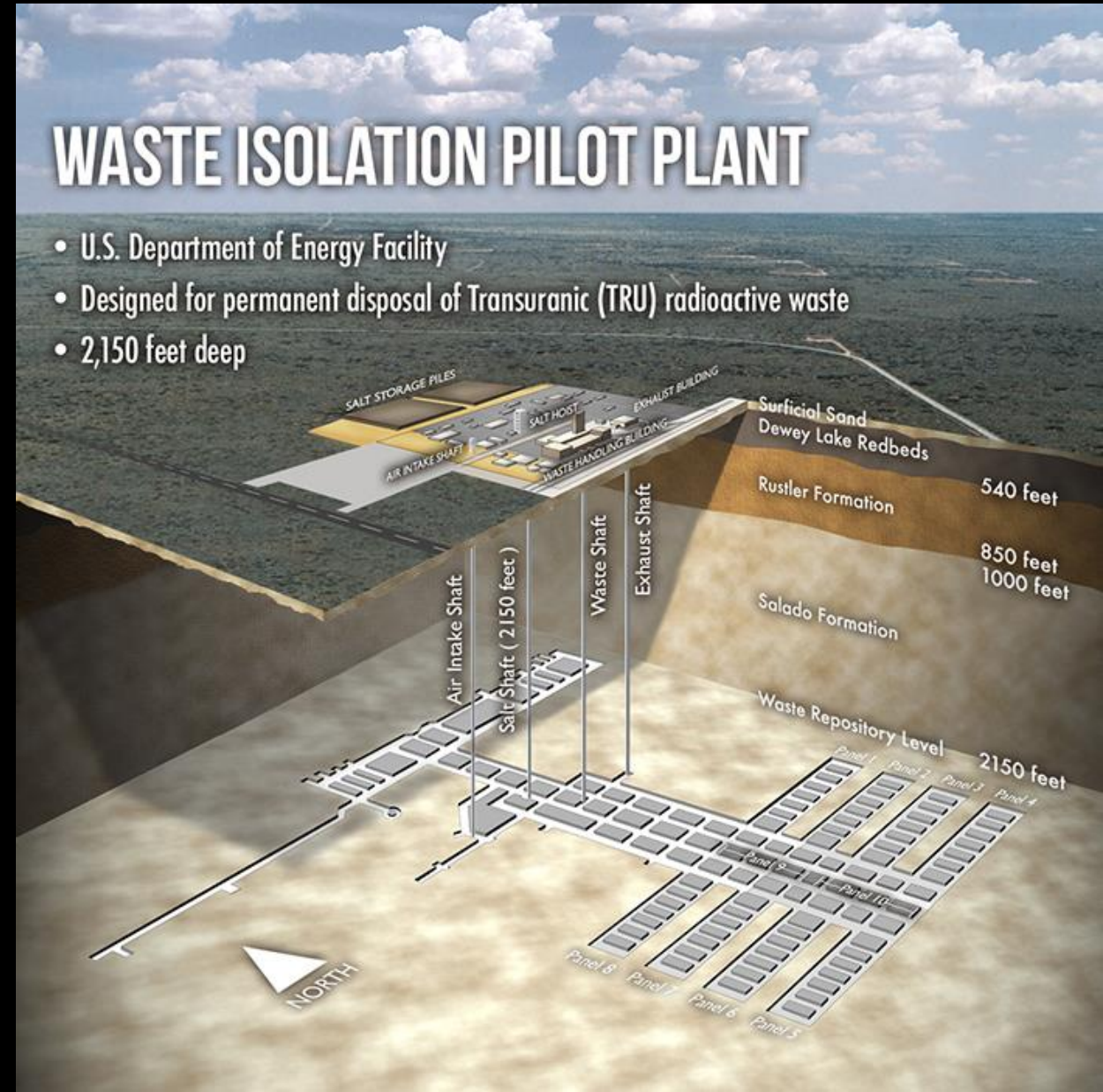
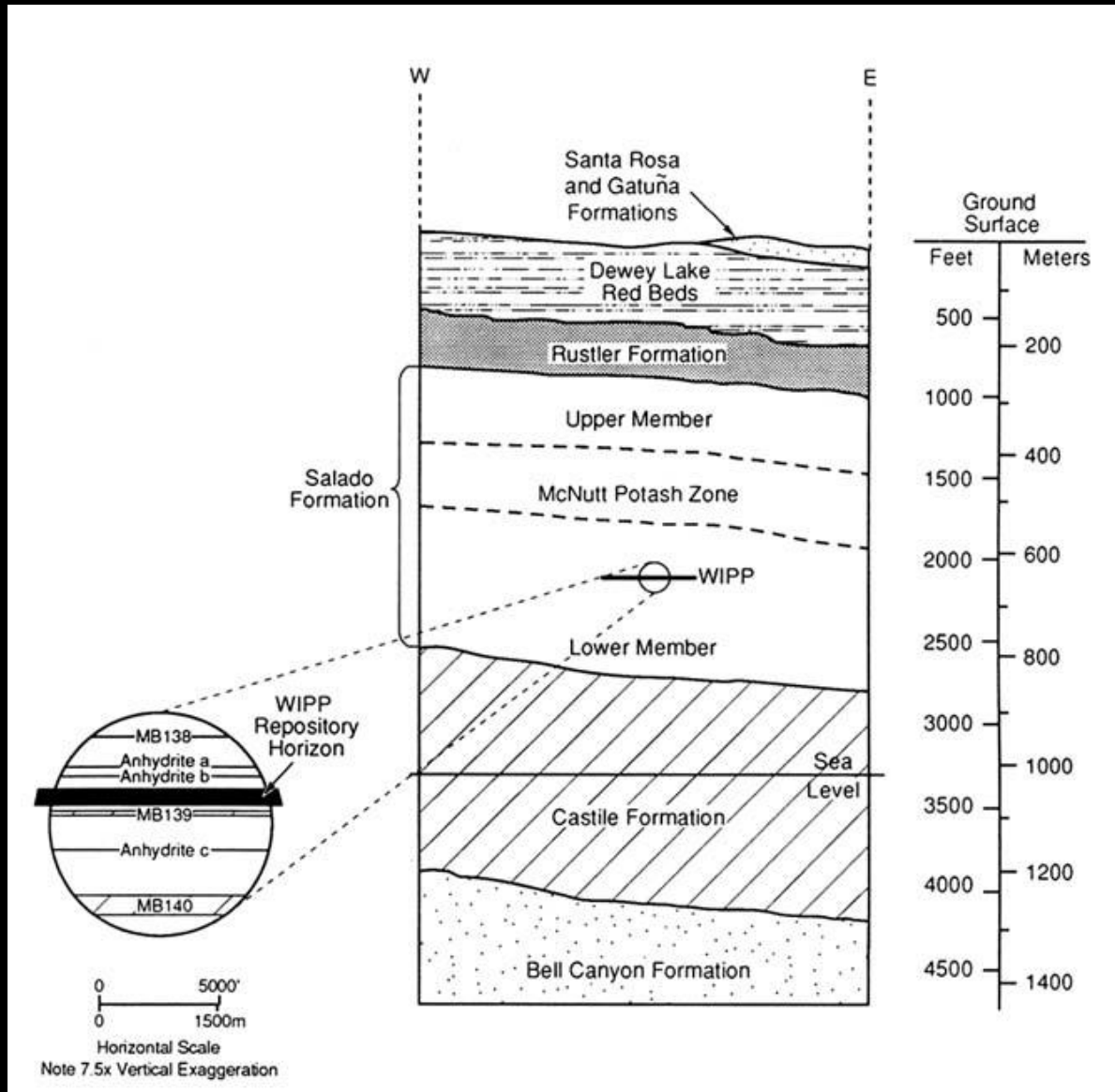


Yucca Mountain

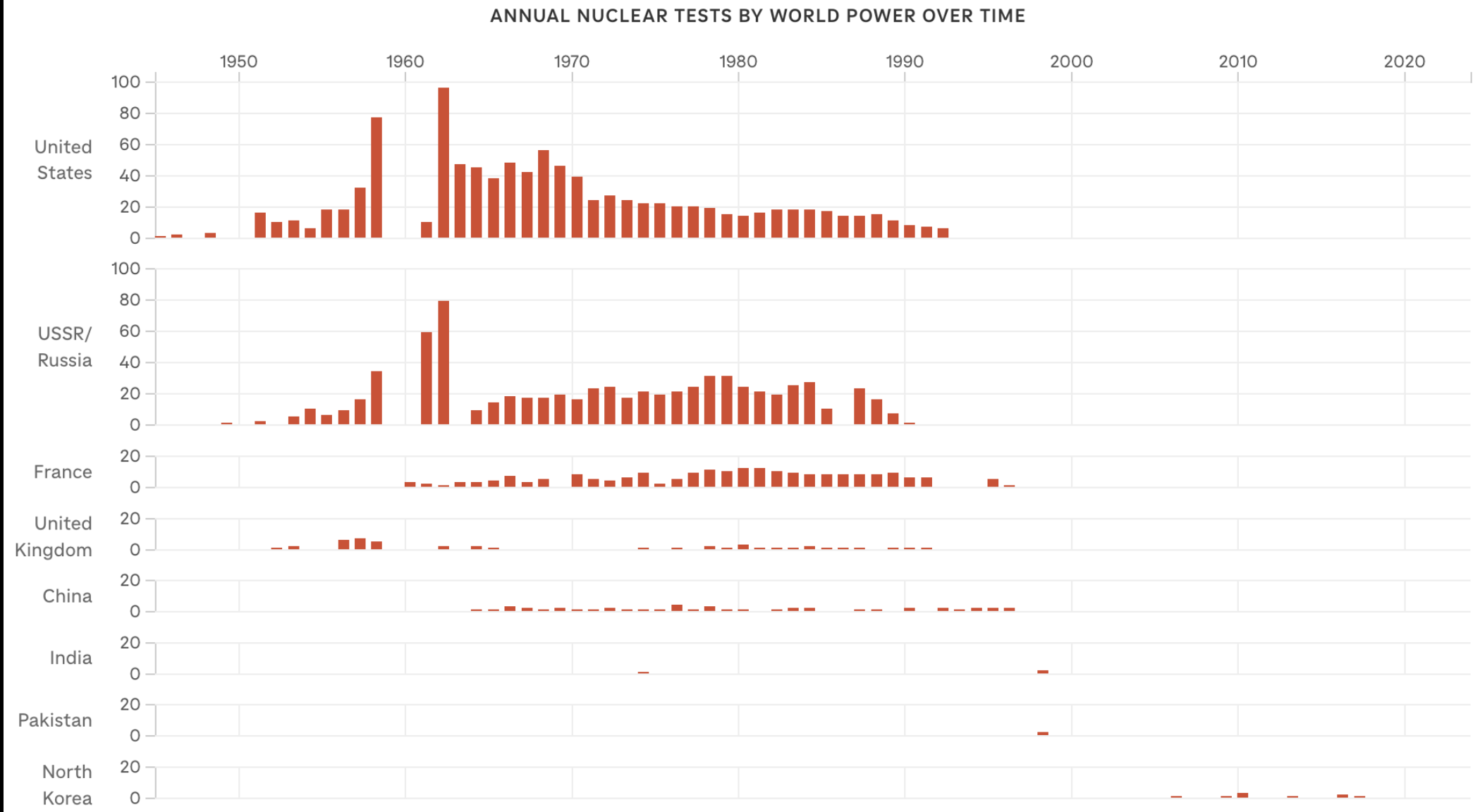


Approved in 2002 as nation's sole deep geologic repository for high-level nuclear waste (both weapons and energy) but ended in 2011 due to extreme opposition in Nevada. Texas passed a law banning disposal in 2021.

Waste Isolation Pilot Plant (WIPP Site)



Nuclear testing largely ended in the 1990s. There are signs that could change



Source: [Arms Control Association](#)

Credit: Brent Jones/NPR

Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO) Monitoring Network



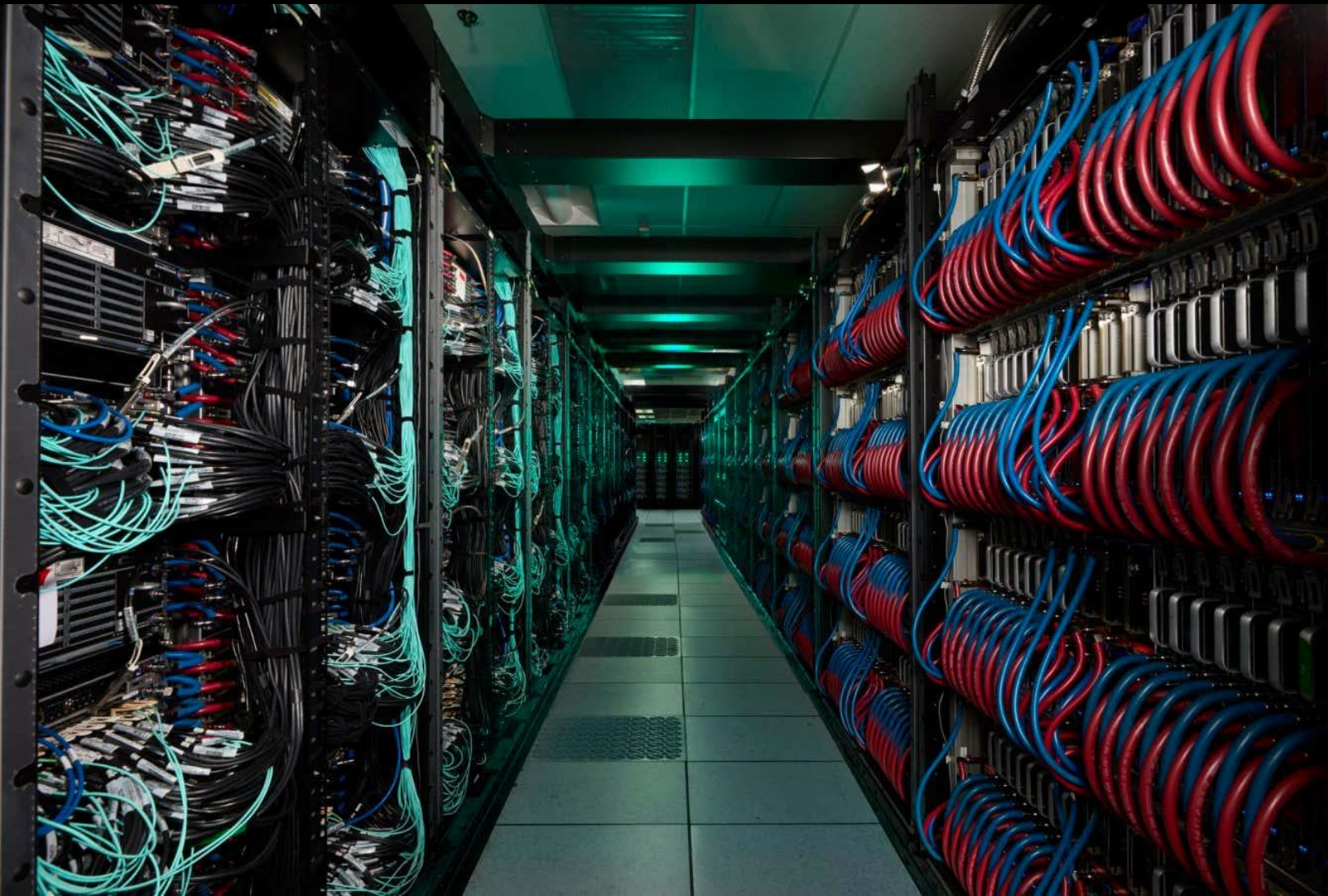
Four complementary technologies

The IMS uses four complementary verification methods, using the latest available technology:

- Fifty primary and 120 auxiliary [seismic stations](#) to monitor for an underground test by measuring shockwaves through the ground.
- Eleven [hydroacoustic stations](#) to detect soundwaves through the ocean from an underwater explosion.
- Sixty [infrasound stations](#) to listen for ultra-low-frequency sound waves moving through the atmosphere at levels inaudible to the human ear.
- Eighty [radionuclide stations](#) to detect radioactive particles or gases from atmospheric explosions, or vented by underground or underwater nuclear explosions. Sixteen radionuclide laboratories assist radionuclide stations in identifying these radioactive substances.

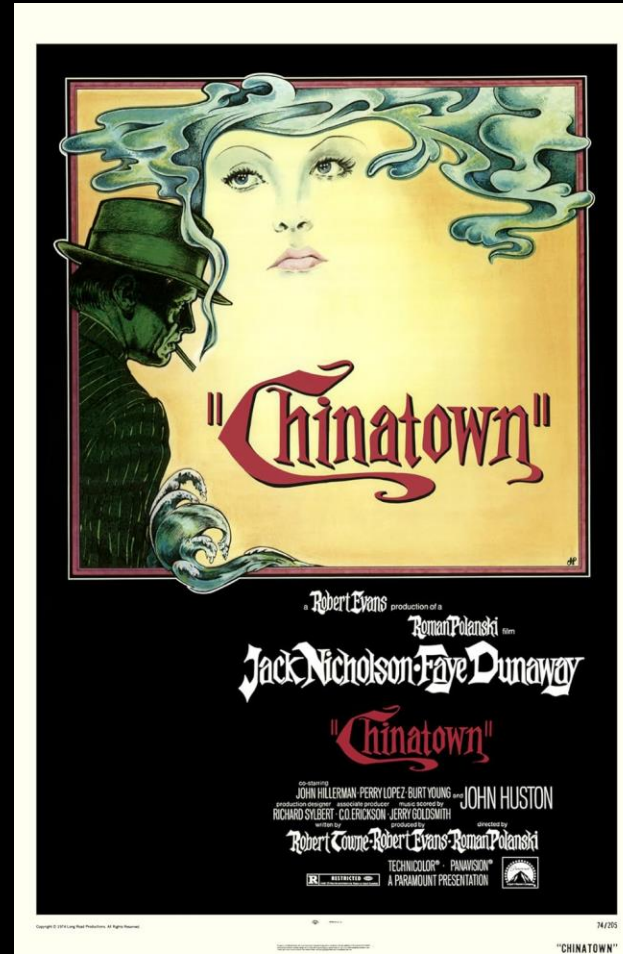
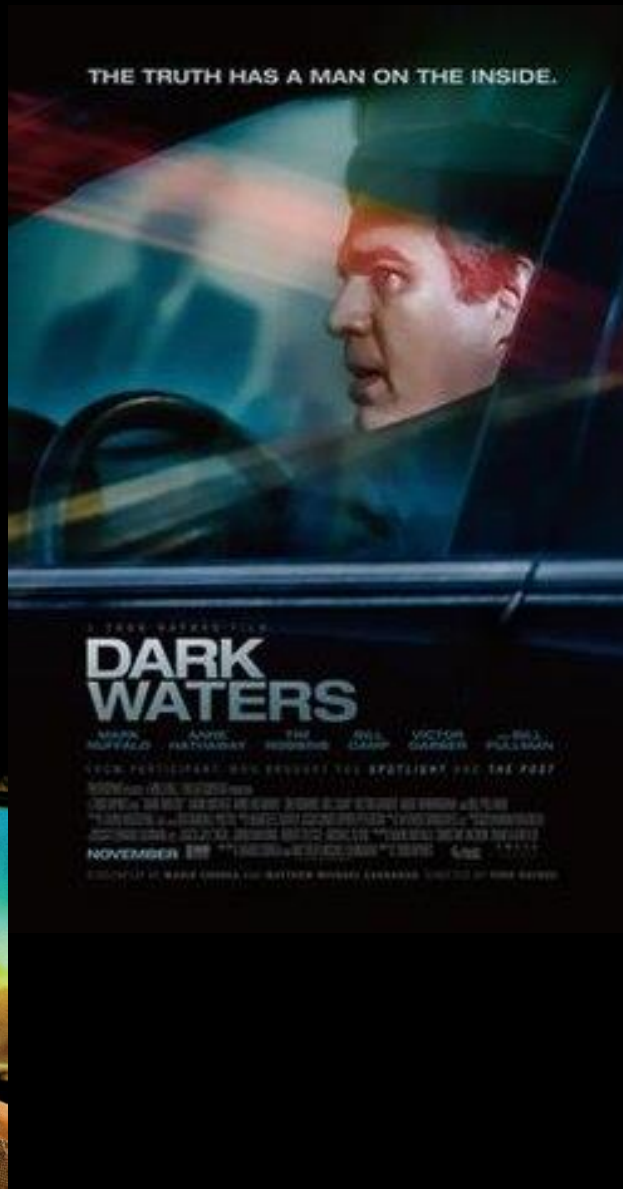
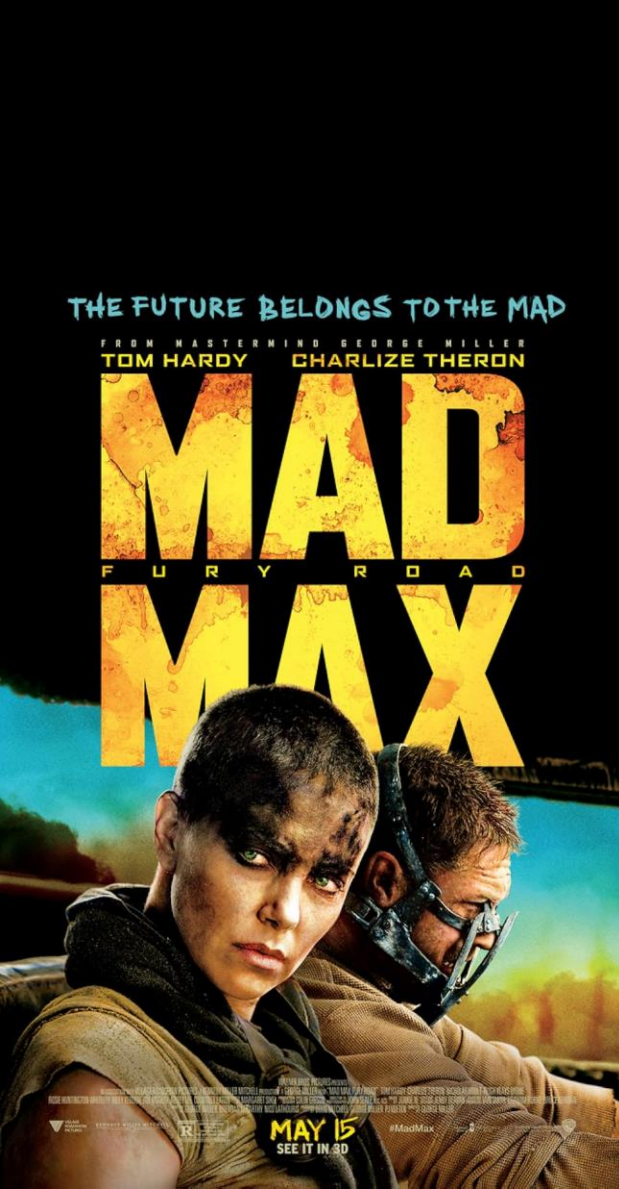
In practice, a single event may be detected by more than one type of monitoring station.

Modern Testing of Nuclear Weapons



National Labs are tasked with maintenance, refurbishment, and testing of aging nuclear stockpile. They do so with “nonnuclear” explosions and sophisticated computer simulations.

Hydrology in Popular Culture



Any others??

Reining in the Rio Grande

~ People, Land, and Water ~



"A CAUTIONARY TALE and an important one, a must-read for students of water politics."—John Nichols, author of *The Milagro Beanfield War*

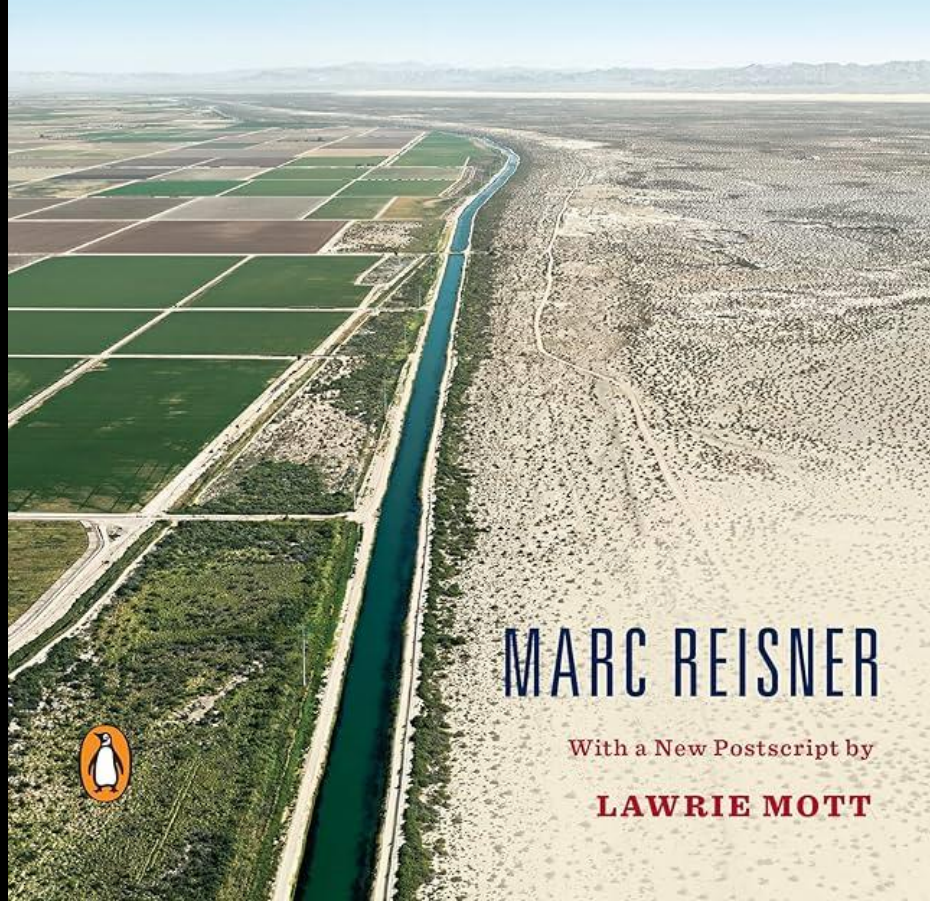


Fred M. Phillips • G. Emlen Hall • Mary E. Black

"The definitive work on the West's water crisis."—Newsweek

CADILLAC DESERT

THE AMERICAN WEST *and its*
DISAPPEARING WATER



MARC REISNER

With a New Postscript by

LAWRIE MOTT



Any others??